

WASP-36b

R-0833

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_230323 · created 2026-07-28T17:52:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460026.669 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.73 +/- 1.59 [%]
Orbital Inclination (inc)	81.57 +/- 2.85
Airmass coefficient 1 (a1)	0.76 +/- 0.23
Airmass coefficient 2 (a2)	0.27 +/- 1.03
Scatter in the residuals of the lightcurve fit is	30.02 %
Best Comparison Star	#2 - [558, 465]
Optimal Method	PSF photometry
Transit Duration (day)	0.06 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.165 ± 0.048 vs 0.1368 (deviation 0.4σ)
Duration fit vs published	0.06 d vs 0.0758 d (deviation 0.38σ)
Mid-time O–C	11.8 min
Depth SNR	2.3
Residual scatter	30.02 %

Light curve

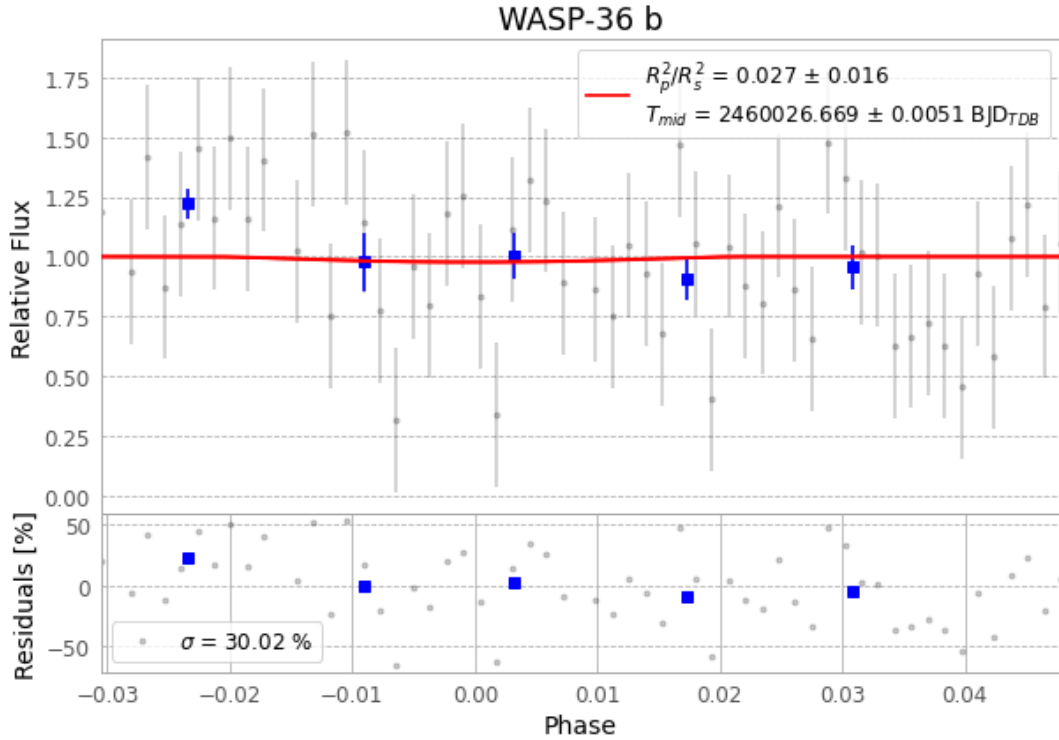


Figure 1 — FinalLightCurve_WASP-36 b_2023-03-22.png (SHA-256 bb19b0e55d7bacf8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [495, 307], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 632f987b2eaab9e8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

58 FITS frames (38 MB), WASP-36230323024846.FITS → WASP-36230323054215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-230323.log (639d199ecf96...), FinalLightCurve_WASP-36 b_2023-03-22.png (bb19b0e55d7b...), FinalLightCurve_WASP-36 b_2023-03-22.csv (8f20efa79195...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 17:52Z.